

Supplemental Online Content

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This supplemental material has been provided by the authors to give readers additional information about their work.

eMethods

Trial Design

The study, registered at ClinicalTrials.gov (NCT04372147), followed the Declaration of Helsinki and Good Clinical Practice guidelines. Ethical approval was obtained from the *Comité de Protection des Personnes (CPP) Ile de France III* on January 10, 2020 for the first version of the study protocol (reference no. 3754- I). All subsequent versions of the protocol, up to the last version (no. 5.0; June 7, 2022) were approved by the same research ethics committee. All protocol amendments were previously reported.¹ All patients or their representatives provided written informed consent.

The trial was designed by the first and two last authors. All investigators and study coauthors vouch for the rigorous fidelity of the trial to the protocol, the exhaustivity and accuracy of data reported, including the reporting of adverse events, and the fidelity of the analysis to the statistical analysis plan. The statistical analysis was performed by a biostatistician of the Clinical Research Unit under supervision of the lead study methodologist. The first draft of the manuscript was written by the first author and revised by the last two authors. The final draft was revised by all authors, who provided agreement for submitting the manuscript for publication.

An independent and blinded committee adjudicated possible recurrence cases to ascertain the primary endpoint. An independent Data Safety Monitoring Board monitored safety data and interpreted the results of the planned interim analysis. The study was sponsored by Assistance Publique Hôpitaux de Paris (AP-HP) and funded by an academic grant from Programme Hospitalier de Recherche Clinique - PHRC 2018 (Ministère de la Santé) (PHRC-IR AOR18104). Merit Medical (South Jordan, UT, USA) and Balt (Montmorency, France)

provided embolization particles, coils and coiling microcatheters free of charge. Terumo Corp. (Shibuya, Tokyo) discounted the price of the Progreat microcatheters. The funders were not involved in the study design, data analysis, interpretation of results or writing or reviewing of trial results or their publication.

Patients

As reported in the main manuscript, Patients were not eligible for inclusion if CSDH evacuation was performed by craniotomy or twist-drill craniostomy rather than burr-hole. As previously reported, openings of the skull up to a diameter of 5 mm were classified as twist-drill craniostomy, openings of up to 30 mm as burr-hole craniostomy, and larger openings as craniotomy.²⁻⁴

Trial Procedures

In the intervention group, patients underwent CT-angiography of the supra-aortic and intracranial vessels followed by MMA embolization within 7 days of surgery, along with best medical treatment. In the third amendment of the protocol (in effect on April 30, 2021), the CT-angiography was no longer mandatory if the patient had CT-angiography scans of the supra-aortic vessels in the past year that the investigator considered of sufficient quality. The CT-angiography was performed to check the feasibility and risk of arterial catheterization and embolization.⁵ Investigators were asked to check specific elements on the CT-angiography including anatomy of the supra-aortic vessels in order to plan for radial or femoral access, the presence of arterial thrombus in the aortic arch precluding aortic arch navigation, and anatomical variants contraindicating MMA embolization (MMA originating from the ophthalmic artery with agenesis of the *foramen spinosum* and absence of an MMA originating from the internal maxillary artery).

The embolization procedure was initially performed under local anesthesia or conscious sedation. Starting from June 10, 2020 (second amendment of the protocol), embolization could also be performed under general anesthesia if necessary. Via a femoral or radial access, the external carotid artery was catheterized with a regular 5F or 6F guiding catheter. Then the MMA was catheterized with a Progreat 2.4 (Terumo, Shibuya, Tokyo) microcatheter in a coaxial fashion above the *foramen spinosum* and embolized by free-flow injection of 300-500 μm EmboSphere (Merit Medical, South Jordan, UT, USA) triascryl gelatin microspheric (TAGM) particles. We chose 300-500 μm TAGM calibrated microspheres as the embolic agent because of previously reported non-randomized data suggesting a potential benefit.⁶ This choice was also guided by safety concerns because these particles are not expected to penetrate the meningeal anastomotic arteries, which have been reported to be 50 to 300 μm in diameter, thus mitigating the potential risk for non-target embolization.^{1,7} If the selective MMA angiogram showed a dangerous anastomosis (an arterial anastomosis exposing to a risk of non-target embolic agent migration towards an eloquent structure such as the eye), the artery was embolized downstream of the anastomosis with a safety margin or the anastomosis was occluded using Optima coils (Balt, Montmorency, France) delivered via a Vasco 10 (Balt) microcatheter. Technically, MMA embolization aimed to occlude both anterior and posterior MMA branches. This was achieved by injection from the common trunk or selective catheterization of each individual branch. Patients with a unilateral CSDH underwent embolization homolaterally to the CSDH. Patients with a bilateral CSDH underwent bilateral embolization. Patients who had unilateral surgery for a bilateral CSDH (i.e., if the CSDH was not compressive on one side and did not require surgery) underwent unilateral embolization on the same side as the surgery. In the specific setting in which a patient had bilateral surgery for a bilateral CSDH, then the CSDH recurred unilaterally and had unilateral surgery for this

recurrence and was included in the study, the embolization procedure was performed bilaterally.

In the control group, patients received the best medical treatment alone, which included standard post-surgical care and nursing care, symptomatic treatments including pain management medication and rehabilitation if necessary. Corticosteroids were authorized if deemed necessary by the attending physician. Clinical and imaging follow-up beyond those planned in the protocol was organized as deemed necessary by the attending physician depending on patient status.

Statistical Analysis

Baseline characteristics are described in each group with median (interquartile range) for quantitative variables and number (percentage) for qualitative variables.

In the primary endpoint analysis, a multiple imputation method was implemented for patients who could not be evaluated for the main endpoint except those who died from a neurological or unknown cause who were considered as having a CSDH recurrence. The Multiple Imputation by Chained Equations (MICE) technique was used to replace missing values.⁸ The variables used in the imputation phase were recurrence at 6 months, age, sex, unilateral or bilateral nature of the CSDH and use of anticoagulant/antiplatelet therapy as well as the characteristics of CSDH at inclusion, namely type of CSDH (initial episode or recurrence), proven recent head trauma, Glasgow Coma Scale (GCS) score, American Society of Anesthesiologists (ASA) score, focal neurological deficit, hemiparesis, phasic disorder, gait disorder, headache, maximum thickness of the CSDH, Park classification⁹, type of surgery (burr hole, trephine craniostomy), complications (infection, subdural empyema, epileptic seizure during surgery or immediately afterward). Analyses involved using R software and

the mice package. Ten datasets were generated. The results from each imputed dataset were pooled using Rubin's rules to obtain overall estimates and their standard errors.

Data Sharing Statement

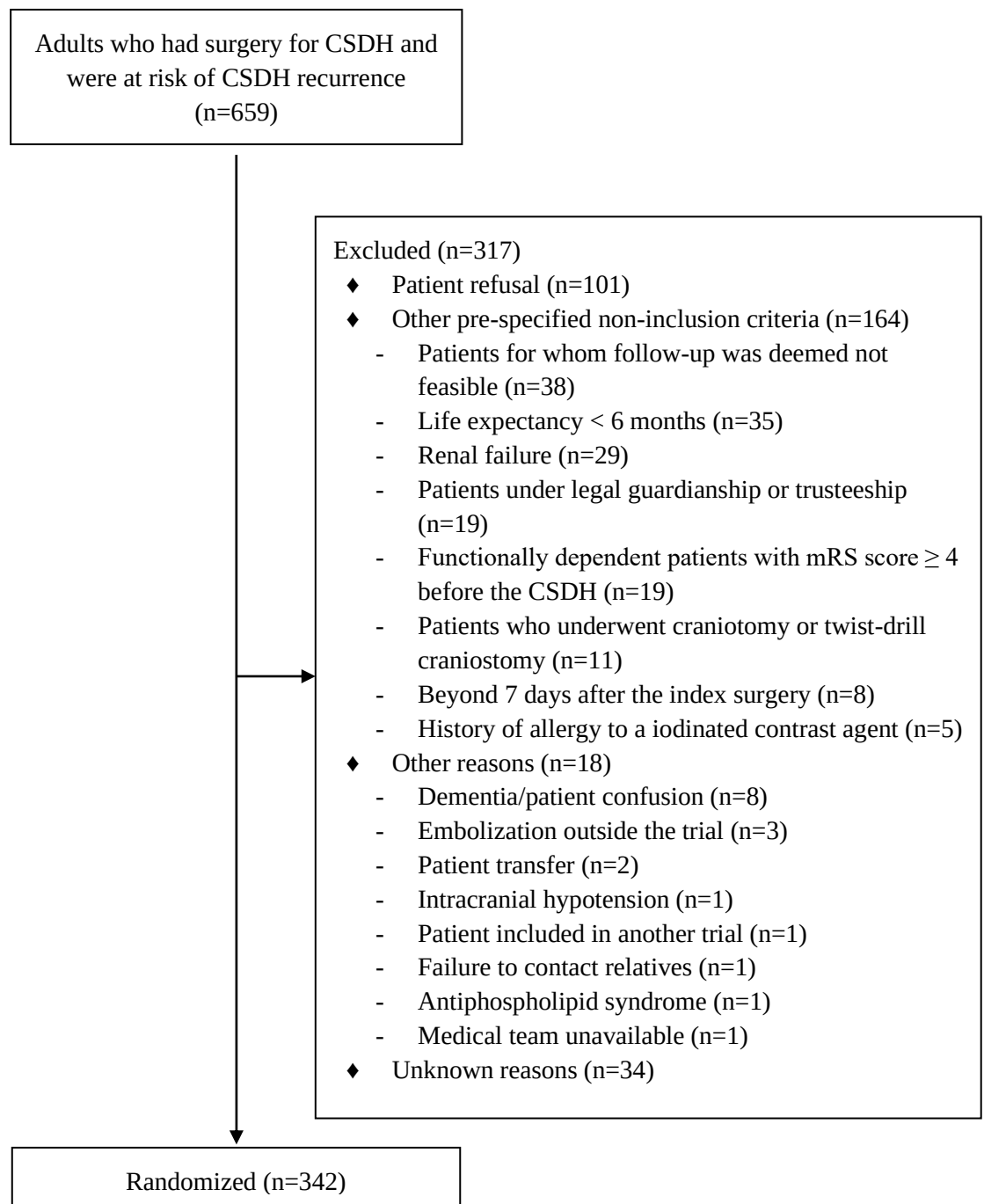
Data are available upon reasonable request by a researcher independent of the study. The procedures carried out with the French data privacy authority (CNIL, Commission nationale de l'informatique et des libertés) do not allow the transmission of the database, nor the information and consent documents signed by the patients. Consultation by the editorial board or interested researchers of individual participant data that underlie the results reported in the article after deidentification may nevertheless be considered, subject to prior determination of the terms and conditions of such consultation and in respect for compliance with the applicable regulations.

eReferences

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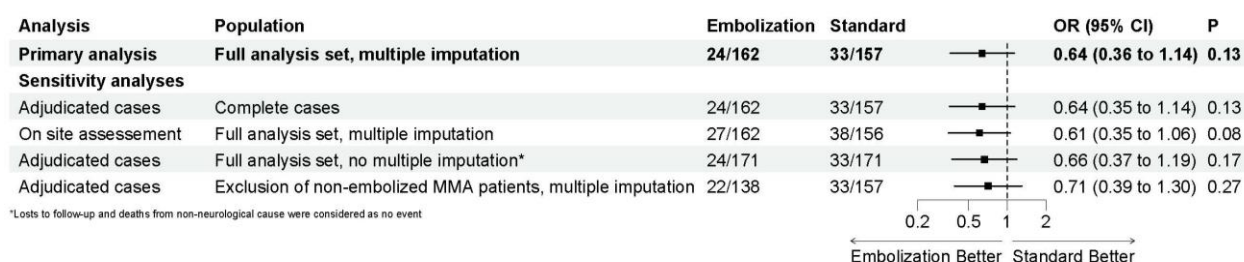
eFigures

eFigure 1. Detailed list of reasons for excluding patients with an inclusion criterion



CSDH denotes chronic subdural hematoma; mRS, modified Rankin Scale.

eFigure 2. Primary endpoint and sensitivity analyses



Forest plot of 6-month CSDH recurrence assessed by the adjudication committee for the primary analysis using a logistic regression model adjusted on randomization stratification factors, namely anticoagulant/antiplatelet use (yes versus no) and CSDH localization (bilateral versus unilateral). The last sensitivity analysis presented excluded patients who did not receive embolization in the experimental group.

CSDH denotes chronic subdural hematoma; OR, odds ratio; CI, confidence interval.

eTables

eTable 1. Distribution of the number of patients randomized by center

Site number	Site name	Embolization N = 171	Standard care alone N = 171	Overall N = 342
001	Paris-APHP - Pitié Salpêtrière	88 / 171 (51.5%)	89 / 171 (52.0%)	177 / 342 (51.8%)
008	Limoges-CHU Dupuytren Limoges	21 / 171 (12.3%)	19 / 171 (11.1%)	40 / 342 (11.7%)
011	Lille-CHRU de Lille - Roger Salengro	19 / 171 (11.1%)	19 / 171 (11.1%)	38 / 342 (11.1%)
012	Tours-CHRU de Tours - Bretonneau	10 / 171 (5.8%)	10 / 171 (5.8%)	20 / 342 (5.8%)
002	Paris-CH Sainte Anne	7 / 171 (4.1%)	10 / 171 (5.8%)	17 / 342 (5.0%)
004	Clichy-APHP - Beaujon	7 / 171 (4.1%)	7 / 171 (4.1%)	14 / 342 (4.1%)
010	Marseille-APHM - Nord	6 / 171 (3.5%)	5 / 171 (2.9%)	11 / 342 (3.2%)
009	Marseille-APHM - Timone Adultes	4 / 171 (2.3%)	5 / 171 (2.9%)	9 / 342 (2.6%)
003	Paris-APHP - Lariboisière	3 / 171 (1.8%)	2 / 171 (1.2%)	5 / 342 (1.5%)
005	Créteil-APHP - Henri Mondor	2 / 171 (1.2%)	2 / 171 (1.2%)	4 / 342 (1.2%)
007	Clamart-HIA Percy	2 / 171 (1.2%)	2 / 171 (1.2%)	4 / 342 (1.2%)
006	Paris-Fondation Ophtalmologique A. de Rothschild	2 / 171 (1.2%)	1 / 171 (0.6%)	3 / 342 (0.9%)

eTable 2. Distribution of event rates for criteria used to define chronic subdural hematoma (CSDH) recurrence at 6 months

*Characteristic	Embolization N = 24	Standard care alone N = 33	Overall N = 57
1. Reappearance of an homolateral CSDH with a midline shift ≥ 5 mm or a symptomatic homolateral CSDH recurrence (including leading to death)	16 / 24 (66.7%)	28 / 33 (84.8%)	44 / 57 (77.2%)
a. Midline shift ≥ 5 mm	11 / 24 (45.8%)	17 / 33 (51.5%)	28 / 57 (49.1%)
b. Symptomatic recurrence, including those leading to death†	15 / 24 (62.5%)	26 / 33 (78.8%)	41 / 57 (71.9%)
2. Homolateral CSDH > 10 mm in maximal thickness on the 6-month control head CT scan	8 / 24 (33.3%)	4 / 33 (12.1%)	12 / 57 (21.1%)
3. Need for repeat surgery for homolateral CSDH recurrence during the study period	7 / 24 (29.2%)	13 / 33 (39.4%)	20 / 57 (35.1%)
4. Need for a new hospital admission related to homolateral CSDH recurrence during the study period)	12 / 24 (50.0%)	15 / 33 (45.5%)	27 / 57 (47.4%)
Death from neurological or undetermined cause considered as CSDH recurrence	2 / 24 (8.3%)	1 / 33 (3.0%)	3 / 57 (5.3%)

* One patient could experience one or more criteria.

†Symptoms were: focal neurological deficit, hemiparesis, speech disorder, gait disorder, headache or intermittent dizziness

eTable 3. List of pre-specified embolization procedure-related complications

Endpoints	Embolization (N = 171)
<i>Major complications</i>	
Major complications – no. (%)	1 (0.6)
Mechanical thrombectomy after the occurrence of intracranial MCA occlusion during carotid catheterization – no. (%)	1 (0.6)
Superficial hematoma with deglobulization (loss of 2 Hb points on CBC) and/or requiring transfusion – no. (%)	0 (0.0)
Retroperitoneal hematoma with or without deglobulization – no. (%)	0 (0.0)
Pseudo arterial aneurysm at the puncture site requiring surgical treatment – no. (%)	0 (0.0)
Occlusion of the femoral artery and/or acute limb ischemia – no. (%)	0 (0.0)
Abscess at the puncture site – no. (%)	0 (0.0)
Anaphylactic shock – no. (%)	0 (0.0)
Quincke's edema – no. (%)	0 (0.0)
Bronchospasm with desaturation – no. (%)	0 (0.0)
Extensive rash – no. (%)	0 (0.0)
Deterioration of renal function with persistent decrease in creatinine clearance of > 10 points – no. (%)	0 (0.0)
Neurological accident with permanent sequelae – no. (%)	0 (0.0)
<i>Minor complications</i>	
Minor complications – no. (%)	3 (1.8)
Superficial hematoma at the puncture site – no. (%)	0 (0.0)

Endpoints	Embolization (N = 171)
Transient deterioration of renal function – no. (%)	0 (0.0)
Transient neurological accident– no. (%)	2 (1.2)
Asymptomatic arterial dissection – no. (%)	0 (0)
Mild headaches – no. (%)	1 (0.6)

MCA denotes middle cerebral artery; Hb, hemoglobin; CBC, complete blood count.

eTable 4. Middle meningeal artery randomized clinical trials summarized

Trial acronym	Funding / Promotion	Setting	Population	Intervention	Control group	Primary endpoint	Primary outcome
EMBOLISE (NCT04402632)	Industry funded by Medtronic	400 patients included in 39 centers in the United States	Symptomatic subacute or CSDH with an indication for surgical evacuation	Ipsilateral or bilateral MMA embolization with a non-adhesive liquid embolic agent (Onyx)	Standard surgical and post-surgical care	Hematoma recurrence or progression that led to repeat surgery within 90 days after the index treatment	8 patients (4.1%) reached the primary outcome in the intervention group as compared to 23 patients (11.3%) in the control group (RR 0.36; 95% confidence interval [CI], 0.11 to 0.80; P = 0.008)
STEM (NCT04410146)	Industry funded by Balt USA	310 included in 32 sites in USA and Europe	Symptomatic CSDH (either surgical or nonsurgical standard treatment chosen for each patient before randomization)	Ipsilateral MMA embolization with a non-adhesive liquid embolic agent (Squid)	Standard surgical and post-surgical care, or medical management, according the patient subgroup	Recurrent or residual CSDH (measuring >10 mm) at 180 days; reoperation or surgical rescue within 180 days; or major disabling stroke, myocardial infarction, or death from neurologic causes within 180 days	19 of 120 patients (16%) reached the primary outcome in the intervention group as compared to 47 of 129 patients (36%) in the control group (OR, 0.36; 95% confidence interval, 0.20 to 0.66; P = 0.001)
MAGIC-MT (NCT04700345)	Co-sponsored by Shanghai Shengkang Hospital Shanghai Municipal Health Commission and Covidien/Medtronic	727 (722 in the intent to treat analysis) included in 31 centers in China	Symptomatic nonacute subdural hematoma with mass effect assigned to burr-hole drainage or nonsurgical treatment at the surgeon’s discretion	MMA embolization with a non-adhesive liquid embolic agent (Onyx) injected ipsilateral the hematoma	Standard surgical and post-surgical care, or medical management, according the patient subgroup	Recurrence defined as a maximum hematoma thickness >10 mm or reoperation during follow-up after previous burr-hole drainage (within 90 days) or progression defined as an increase in the maximum hematoma thickness >3 mm from baseline or surgical rescue in patients who were receiving usual care (within 90 days)	24 patients (6.7%) reached the primary outcome in the intervention group as compared to (9.9%) in the usual-care group (between-group difference, −3.3 percentage points; 95% confidence interval, −7.4 to 0.8; P = 0.10)

EMPROTECT (NCT04372147)	Publicly funded and promoted by "Assistance Publique Hopitaux de Paris" with industrial support (Merit Medical, Balt, Terumo)	342 patients included in 12 centers in France	Patients having undergone surgery for CSDH recurrence or surgery for a first CSDH episode at high risk of recurrence	MMA embolization 300-500µm calibrated TAG microspheres (Embosphere) injected ipsilateral to the CSDH	Standard surgical and post-surgical care	CSDH recurrence rate at 6 month: reappearance of an homolateral CSDH with a midline shift ≥ 5 mm or a symptomatic homolateral CSDH, including leading to death; or the presence of a homolateral CSDH > 10 mm in maximal thickness (6-month control head CT scan); or repeat surgery for homolateral CSDH recurrence; or a new hospital admission related to homolateral CSDH recurrence.	24/162 (14.8%) patients reached the primary outcome in the intervention group as compared to 33/157 (21.0%) patients in the control group (ITT with imputation: OR, 0.64; 95% confidence interval, 0.36 to 1.14; p=0.13)
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CSDH denotes chronic subdural hematoma; MMA, middle meningeal artery.